

Pandas Cheatsheet — Essential Functions (A4)

Data Loading & Inspection	
pd.read_csv("file.csv")	Load CSV
pd.read_json("data.json")	Load JSON
pd.read_excel("file.xlsx")	Load Excel
df.info()	Schema + nulls
df.head(10)	Preview rows
df.tail(5)	Last rows
df.sample(5)	Random sample
df.shape	Rows/columns
df.dtypes	Column types
df.columns	Column names
df.describe()	Summary stats

Cleaning & Transforming	
df.dropna(subset=['id'])	Drop missing
df.dropna(how='all')	Drop all-null rows
df.dropna(inplace=True)	Drop missing in-place
df.fillna(0)	Fill missing
df.fillna(0, inplace=True)	Fill in-place
df.ffill()	Forward fill
df.drop_duplicates(subset=['id'])	Deduplicate
df.duplicated()	Check duplicates
df.columns.difference(['id'])	Exclude columns
df.reset_index(drop=True)	Reset index
df.reset_index(drop=True, inplace=True)	Reset index in-place
df.rename(columns={'a': 'A'})	Rename
df.rename(columns={'a': 'A'}, inplace=True)	Rename in-place
df.drop(columns=['col'])	Drop column
df.drop(columns=['col'], inplace=True)	Drop column in-place
df.replace({'A': 1})	Replace values
df.assign(new_col=df['x']*2)	Add column
np.where(df['x']>0, 1, 0)	Conditional column

Advanced — Window & Cumulative	
df['ma7'] = df['sales'].rolling(7).mean()	Rolling mean
df['lag'] = df['x'].shift(1)	Lag column
df['r'] = df.groupby('cat')['sales'].rank()	Rank
df.groupby('cat').cumcount()	Cumcount per group
df['x'].cumsum()	Cumulative sum
df.groupby('cat')['x'].cumsum()	Cumsum per group
df['x'].cummax()	Cumulative max
df['x'].cummin()	Cumulative min
df['x'].pct_change()	Pct change
df['x'].diff(1)	Diff (lag diff)

Selecting & Filtering	
df['col']	Select column
df[['a', 'b']]	Select multiple columns
df.iloc[0:5]	Select by position
df.loc[df['x']>0]	Select by label/condition
df[df['age'] > 30]	Boolean filter
df[(df['age']>18) & (df['country']=='US')]	Multi-condition
df.query("country=='US'")	SQL-style filter
df[df['age'].between(20, 30)]	Range filter
df[df['id'].isin([1, 2, 3])]	Is-in filter

Aggregation & Grouping	
df.groupby('cat')['sales'].sum()	Single col group
df.groupby(['cat', 'region'])['sales'].sum()	Multi-col group
df.groupby(['cat', 'region']).agg({...})	Multi-col + multi-agg
df.groupby('cat').mean()	Group mean
df.groupby('id').agg({'a': 'sum', 'b': 'count'})	Multi-agg
df.groupby('cat').size()	Group size
df['id'].count()	Count
df['user_id'].nunique()	Unique count
df['country'].value_counts()	Frequency
df.sort_values('sales', ascending=False)	Sort
df.sort_values('sales', inplace=True)	Sort in-place
df.nlargest(5, 'sales')	Top N rows

Type Handling	
df['id'].astype('Int64')	Cast type
df['id'].astype(str)	To string
pd.to_datetime(df['ts'])	To datetime
df['ts'].dt.date	Extract date
df['ts'].dt.year	Extract year
df['ts'].dt.month	Extract month
df['ts'].dt.dayofweek	Day of week

Joining & Combining	
pd.merge(a, b, on='id', how='left')	Left join
pd.merge(a, b, on='id', how='right')	Right join
pd.merge(a, b, on='id', how='inner')	Inner join
pd.merge(a, b, on='id', how='outer')	Outer join
pd.merge(a, b, how='cross')	Cross join
df1.join(df2, how='left')	Index join
pd.concat([df1, df2])	Stack DataFrames
pd.concat([df1, df2], axis=1)	Concat columns

Advanced — Reshape & Strings	
df.pivot_table(values='sales', index='cat', columns='region')	Pivot
df.melt(id_vars=['id'])	Wide to long
df.explode('col')	Explode list col
df['x2'] = df['x'].apply(lambda x: x*2)	Apply
df['email'].str.lower()	String lower
df['col'].str.contains('x')	String contains
df['col'].str.split(',')	String split
df.loc[df['age']<0, 'age'] = 0	Update rows